



Temporal assessment of N-cycling genes and nutrient-acquiring enzymes under rotational grazing

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Received: 6 January 2026 / Accepted: 7 September 2026
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Abstract Despite the importance of microbial activity for the sustainability of grazing systems, longitudinal assessments of microbial responses to grazing remain scarce. We monitored functional N-cycling genes and C-, N-, and P-acquiring extracellular enzyme activities across two contrasting systems: Native Rangeland (mixed prairie vegetation) and Introduced Pasture [*Cynodon dactylon* (L.) Pers.]. Two grazing strategies were compared:

Adaptive (AD; higher stocking density with variable rest periods) and Prescriptive (PR; lower stocking density with fixed rest periods) across three years. Strong temporal variation was evident across genes and enzyme activities, with greater values in 2022 compared to 2023 and 2024. Overall, grazing affected the nitrate reduction and denitrification pathways. In Native Rangeland, the abundance of *napA* was up to 57% greater under AD, *narG* up to twofold greater under PR, and *nirS* showing inconsistent grazing treatment effects. In Introduced Pastures, *nirS* was up to 57% greater under PR, whereas *nirK* showed inconsistent responses. Grazing greatly affected enzyme activity in Native Rangeland under AD, with C-enzymes up to 175%, N-enzymes up to 271%, and P-enzymes up to 216% compared to PR. In Introduced Pastures the greatest differentiation was for N-enzymes, with PR exhibiting two-fold the activity of AD. Overall, our results show that grazing mediates microbial nutrient acquisition in a context-dependent manner. These findings reinforce the importance of accounting for site-specific conditions when applying grazing strategies and interpreting soil microbial responses.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10705-026-10526-9>.

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Keywords Native rangeland · Adaptive multi-paddock grazing · Soil N pathways · Nutrient acquisition enzymes

Introduction

Grazing management is a key driver of soil microbial functioning in grassland ecosystems because it regulates plant biomass removal (Souther et al. 2019), animal-derived nutrient returns (Bai et al. 2012; Abagandura et al. 2024), and soil physical disturbance (Pereira Valani et al. 2025). Grazing management is characterized by factors such as stocking method, stocking density, grazing frequency, and rest period, which influence microbial activity by controlling timing and intensity of defoliation, trampling, and nutrient inputs to the soil (Van Syoc et al. 2022; Feng et al. 2025). Unlike continuous or prescriptive grazing methods that follows a pre-established calendar rotation, adaptive multi-paddock (AMP) grazing involves moving the herd between paddocks according to forage growth and animal nutritional needs (Teague et al. 2013; Mehre et al. 2024). Additionally, AMP may involve greater stocking density during a shorter period (Van Syoc et al. 2022).

Overall, grazing effects on microbial activity may occur through direct pathways such as trampling, defoliation, and excreta deposition (Yao et al. 2023), as well as indirect pathways mediated by changes in plant community composition (Sparks et al. 2026) and soil structure (Byrnes et al. 2018). These direct and indirect pathways can alter substrate availability and soil microenvironments, ultimately shaping the microbial community composition and activity (Van Syoc et al. 2022; Taylor et al. 2025; Meléndez-Aldana et al. 2025). Previous studies showed that AMP grazing increased soil microbial populations and nutrient cycling (White et al. 2023) and soil organic C stocks (Mehre et al. 2024) in comparison to conventional (low to moderate stocking densities with infrequent cattle movement) or continuous grazing.

Grazing-induced changes in soil microbial communities are associated with shifts in the abundance of functional genes, resulting in increased N mineralization and nitrification while decreasing denitrification compared to non-grazed systems (Yang et al. 2013). Similarly, Liu et al. (2023) compared non-grazed and grazed soils and reported that grazing-induced changes in microbial functional genes abundance can alter soil P solubilization and mineralization. Although gene abundance may not directly reflect potential enzyme activity, as only a portion of the genetic pool is actively expressed at a given time

(Enwall et al. 2010), soil microbial functional genes are highly sensitive to management (Jia et al. 2025; Li et al. 2025). These studies showed that soil microbial functional genes can provide a more detailed and mechanistic understanding of soil processes and microbial functioning than traditional biochemical indicators alone. While functional genes indicate the potential for microbial transformations, extracellular enzymes provide a complementary measure of microbial nutrient acquisition (Ndabankulu et al. 2022; Daunoras et al. 2024). Because these enzymes respond directly to microbial metabolic requirements, their activity and stoichiometry provide valuable insights into soil fertility, nutrient limitations, and overall microbial functioning (Daunoras et al. 2024; Morales-Salmerón et al. 2025).

Grazing represents a major disturbance in grassland ecosystems and can alter enzyme activity, although the magnitude and direction of these effects often depend on vegetation characteristics, including growth patterns and photosynthetic pathways (Wang et al. 2023a; Feng et al. 2025). Overall, heavy grazing reduces plant biomass and litter input, thereby limiting nutrient recycling and microbial nutrient supply, which can deplete soil C, N, and P pools and restrict the substrates available for enzyme production (Ji et al. 2014; Tang et al. 2019; Bai et al. 2021; Feng et al. 2025). Consequently, enzyme activities often decline with increasing grazing intensity; however, long-term grazing may lead to microbial and enzymatic adaptations that partially offset these negative effects over time (Feng et al. 2025). Although the suitability of extracellular enzyme activity measurements as proxies for overall microbial activity remains debated, these assays serve as useful early indicators of shifts in microbial nutrient cycling in response to grazing management (Luo et al. 2017; Zhang et al. 2024; Zhao et al. 2025).

Although grazing can have long-term effects on microbial functioning, climatic factors are the major drivers of seasonal variation in soil microbial activity and nutrient cycling (Bai et al. 2021). Seasonal shifts in soil temperature and moisture alter microbial metabolism, growth rates, and substrate availability, thereby influencing the rates of key biogeochemical processes such as decomposition, nutrient mineralization, and extracellular enzyme production (Szukics et al. 2010; Waghmode et al. 2018; Bytnerowicz et al. 2022; Zhao et al. 2023; Daunoras et al. 2024).

Grazing can further interact with these climate conditions to differentially affect extracellular enzyme activity and, consequently, SOM decomposition (Rajper et al. 2024). Given these interacting temporal and management effects, longitudinal studies, which involve observations over multiple time points, are particularly valuable for understanding how grazing alters microbial functioning within dynamic environmental contexts (Van Syoc et al. 2022; Li et al. 2022; Bastani et al. 2023; Xu et al. 2025). This 3-yr study evaluated the effect of different grazing strategies on N-cycling genes abundance and enzyme activities in both native and introduced systems in the southern Great Plains. We hypothesized that greater stocking densities, followed by longer rest periods under adaptive grazing would alter microbial functioning relative to prescriptive grazing management, resulting in distinct patterns of N-cycling genes abundance and extracellular enzyme activities across grazing treatments. Understanding how grazing affects microbial functionality, while accounting for seasonal variability, can provide a better understanding of its impact on soil health, nutrient cycling, and overall ecosystem functioning. Because the studied systems differ in soil, vegetation, and land-use history, grazing effects were evaluated within each system independently.

Material and methods

Study area

This study was conducted on two ranches at the Noble Research Institute, in the southern Oklahoma—USA: Coffey Ranch (33.9384° N, −97.219147° W) and Red River Grazing Facility (33.896825° N, −97.272948° W). The climate is classified as Cfa according to the Köppen climate classification system (Kottek et al. 2006), representing a humid subtropical condition, characterized by hot and humid summers, and cool to mild winters. Historical mean annual temperature is 17 °C, with a bimodal rainfall distribution peaking in May–June, and September–October, with an annual accumulated precipitation of 967 mm (Fiebrich and Illston, 2025). According to NOAA (2021), precipitation occurs on approximately 68 days per year, with the greatest frequency of precipitation during spring (21 days) and Summer (19 days), followed by Autumn (15 days) and Winter (13 days).

Coffey Ranch is a native prairie rangeland with a mixed plant community. Common species include little bluestem [*Schizachyrium scoparium* (Michx.) Nash], Indiangrass [*Sorghastrum nutans* (L.) Nash], broomsedge bluestem (*Andropogon virginicus* L.), western ragweed (*Ambrosia psilostachya* DC.), purpletop [*Tridens flavus* (L.) Hitchc.], and white heath aster [*Symphyotrichum ericoides* (L.) G.L. Nelson]. In contrast, Red River Grazing Facility was formerly cropland and has since been converted to an introduced pasture system. From 1980 to 2017, this site was intensively cropped with peanuts (*Arachis hypogaea* L.) and cereal rye (*Secale cereale* L.), including conventional use of tillage, fertilization, pesticide application, and irrigation. In 2017, the site was planted with bermudagrass [*Cynodon dactylon* (L.) Pers.] for hay production and continuously fertilized and pesticide treated until 2020. From late 2020 to until June 2022, this site was transitioned from hay production to being directly grazed. During this time, cover crops were introduced, pastures were rotationally grazed, and fertilizer and pesticide use was reduced but not eliminated. After June 2022, fertilizer and pesticide applications were discontinued, distinguishing current management from typical bermudagrass production systems. Predominant plant species are bermudagrass [*Cynodon dactylon* (L.) Pers.], along with spring annuals such as vetch (*Vicia* spp.) and cereal rye (*Secale cereale* L.). Common native species include sixweeks fescue [*Vulpia octoflora* (Walter) Rydb.], maretail [*Conyza canadensis* (L.) Cronquist] and tropic croton (*Croton glandulosus* L.). The soils are classified as Mollisols; at Coffey Ranch, the predominant soil series include Durant, Scullin, Slaughterville, and Teller, whereas at the Red River Grazing Facility, the Slaughterville series is the predominant according to USDA Soil Taxonomy (Soil Survey Staff, 2025). To better represent the contrast between the ranches, henceforth, Coffey Ranch will be referenced as “Native Rangeland”, and Red River Grazing Facility will be “Introduced Pasture”. More details of the vegetation and management history of both ranches can be found in Sparks et al. (2026).

Experimental design

Six pastures (averaging 9.7–12.4 ha each) were selected to be monitored over time on each ranch. The experiment was conducted using a randomized

complete block design with three replicate blocks based on ecological site descriptions to account for potential variability across experimental units. Each block was a split-plot design where the grazing treatments were randomly assigned. Two grazing treatments were implemented: an adaptive (AD) and a prescriptive (PR) approach, designed to evaluate flexible versus fixed rotational management. AD is characterized by high density short-duration stocking [similar to what was described by Allen et al. (2011) as mob stocking, or AMP as described by Teague and Kreuter (2020)] associated with variable rest periods. PR grazing is characterized by low density medium-duration stocking and fixed rest period during the growing season, usually between April and November (75-days rest period for Native Rangeland and 60-days for Introduced Pasture). Grazing days varied based on forage availability and season. When forage availability was less than the target ($1,680 \text{ kg ha}^{-1}$ of dry matter), then the plots would be deferred beyond the usual rest period (usually during winter season), until they could be grazed again in spring of the following year. The stocking density was calculated for each grazing event, following Allen et al. (2011).

In each ranch, both AD and PR were grazed by two different herds of Angus-cross cow-calf pairs composed of the same number of animals and similar live-weights. The AD plots were subdivided into smaller paddocks to allow for the use of greater stocking densities, and cattle were moved daily, with the target of achieving a minimum stocking density ten-fold greater than PR. Both grazing treatments had a target of 50% forage utilization and at least $1,100 \text{ kg ha}^{-1}$ of dry matter target residual biomass. The herds grazed additional areas of the ranches while not in the study areas.

This study was performed from June 2022 through December 2024. To standardize soil assessments over time, three permanent sampling areas (i.e. triangle; 140 m^2) were established in each pasture prior to the grazing initiation and maintained for the duration of the study.

Soil sampling and analysis

Soil samples (0–15 cm depth) were collected with a 1.9-cm diameter soil probe (JMC Soil Samplers, Iowa, USA) following the grazing schedule: in

Native Rangeland, with slower forage recovery rates, samples were collected 30 days after the grazing event (4 replicates per triangle, totaling 12 samples per pasture); in Introduced Pasture, samples were collected 15 days after the grazing event (2 replicates per triangle, totaling 6 samples per pasture). Each soil sample was composed of nine soil cores sub-sampled randomly within the triangles and stored at 4°C until processing. Samples were then homogenized, sieved, and segregated into two portions for further analyses: soil characterization and enzyme activity (kept at 4°C) and DNA extraction (kept at -80°C) following sample handling and pretreatment guidelines (Dick 2011). Samplings started in June 2022 (summer) and occurred three times per year during the growing season [spring (March–May), summer (June–August), and fall (September–November)] through 2024, totaling eight sampling events.

For soil characterization, soil samples were sent to Regen Ag Lab, LLC (Pleasanton, Nebraska, US) where the sample was air-dried and sieved (2 mm). Soil pH was determined using the potentiometric method (soil–water ratio 1:1) (Watson and Brown 1988) and soil organic matter (SOM) by loss of ignition (Combs and Nathan 1998). For soil respiration, samples were incubated for 24 h at 24°C , and the carbon dioxide produced was quantified in SR-1 Soil Respiration System created by Soil Health Innovations, LLC [Pleasanton, NE, USA (Haney et al. 2008)]. Water extractable organic carbon, and organic nitrogen were determined in elemental analyzer Vario TOC-TN (Haney et al. 1999), then used to calculate soil dissolved organic C to dissolved organic N ratio (C:N). Nitrate and ammonium were extracted with potassium chloride and determined colorimetrically (Haney et al. 2006). Soil samples were also treated with organic acids (malic, citric, and oxalic). From the slurry, total phosphorus was determined using Inductively Coupled Plasma spectroscopy, inorganic phosphorus (IP) was determined colorimetrically, and organic phosphorus (OP) was calculated by subtracting the inorganic fraction from total phosphorus (Haney et al. 2006). Soil gravimetric water content (GWC) was determined using the drying method (105°C , 24 h). The descriptive analysis of the soil parameters at the beginning of the study can be found in Table S1.

DNA extraction and N-cycling genes determination

Soil DNA was extracted from 500 mg of soil using the DNeasy PowerSoil Pro Kit (Qiagen, Germantown, Maryland, USA). The extracted DNA was quantified using the Qubit dsDNA HS and BR Assay Kits (Invitrogen, Waltham, Massachusetts, USA). Quantitative PCR (qPCR) was performed to quantify levels of N cycling genes, using 16S and ITS primers (Table S2). Reactions were performed in a 10 µl volume containing 2 µM of each primer, approximately 2 µg of total soil DNA (3 µl) and 5 µl of 2× Forget-Me-Not™ EvaGreen® qPCR Master Mix (Biotum, Fremont, CA). qPCR was run on the QuantStudio 7 flex System (Applied Biosystems, Foster City, CA). Amplification specificity was checked with a dissociation curve acquired by heating the samples from 60 to 95 °C. Normalization and comparison of mean threshold cycle (Ct) values were performed as described by (Livak and Schmittgen 2001). To compare relative abundance of target genes, the mean Ct of triplicate reactions was normalized by that of 16S gene. The selected N-cycling genes are involved in the process of N fixation (*nifH*), ammonia oxidation (*amoA*), and denitrification (*napA*, *narG*, *nirK*, *nirS*, and *nosZ*).

Extracellular enzyme activities and stoichiometry

C, N, and P cycling extracellular enzymes were quantified fluorometrically following Bell et al. (2013) and German et al. (2011). Soil samples were collected and kept at 4 °C for 15 days prior to analysis. Briefly, 2 g of field-moist soil was blended with 50 mL of 50 mM sodium acetate at pH 6 (average pH of all soil samples). The suspensions (soil slurry + substrate) were incubated at 250 rpm at 20 °C for four hours. Following incubation, 20 µL of 0.5 M NaOH was added to the plates. Fluorescent assays of α-glucosidase (AG), β-glucosidase (BG), β-D-cellubiosidase (CB), β-xylosidase (XYL), leucine aminopeptidase (LAP), N-acetyl-β-glucosaminidase (NAG), and phosphatase (PHOS) were read at an excitation wavelength of 360 nm and an emission wavelength of 450 nm using a BioTek Synergy HTX reader (BioTek, Inc., Winooski, VT, USA).

The above-listed enzymes are involved in reactions that produce assimilable C and nutrients from C-, N-, and P-rich organic matter (Sinsabaugh et al. 2008), which are considered as appropriate indicators

of overall C, N, and P acquisition (Wang et al. 2023a, b). Therefore, C-enzymes comprise the sum of AG, BG, CB, and XYL activities; N-enzymes the sum of LAP and NAG activities; and PHOS represents the P-enzymes (Cui et al. 2022; Liu et al. 2020; Massaccesi et al. 2023; Sinsabaugh and Shah 2011; Zhang et al. 2024; Zhao et al. 2025). The enzyme stoichiometry was calculated using enzymatic ratios (Eqs. 1–3), as an indication of enzymatic resources directed towards acquisition of organic N and organic P relative to C, and organic P relative to N (Sinsabaugh et al. 2008):

$$C : N = \frac{\ln(AG + BG + CB + XYL)}{\ln(LAP + NAG)} \quad (1)$$

$$C : P = \frac{\ln(AG + BG + CB + XYL)}{\ln PHOS} \quad (2)$$

$$N : P = \frac{\ln(LAP + NAG)}{\ln PHOS} \quad (3)$$

where $\ln(AG + BG + CB + XYL)$ is the natural log of the sum of C-acquiring enzymes, $\ln(LAP + NAG)$ is the natural log of the sum of N-acquiring enzymes, and $\ln PHOS$ is the natural log of the P-acquiring enzyme. According to Sinsabaugh et al. (2008), C:N:P ratio of enzyme activities remains near 1:1:1, therefore, a value of 1 can be used as the threshold for identifying potential nutrient limitations.

Data analysis

All statistical analyses were conducted using R software version 4.2.1 (R Core Team 2022). Potential outliers were first assessed using boxplots grouped by grazing treatment and sampling time. Outliers were identified within each grazing treatment–time combination using the Tukey interquartile range (IQR) method and removed prior to analysis. Data distribution was inspected visually through histograms and Q–Q plots and statistically evaluated with the Shapiro–Wilk test. Generalized linear mixed models (GLMMs) were fitted for each variable using Gamma family [*glmer* function from the “lme4” package (Bates et al. 2015)]. Fixed effects included grazing, season, year, and their interactions. Gravimetric water content was used as a covariate. Random effects accounted for nested experimental structure

and repeated measures, with sampling locations nested within blocks. Model residuals were evaluated for normality (Q–Q plots of Pearson residuals), homoscedasticity (Pearson residuals vs. fitted values), and overdispersion. When applicable ($p < 0.05$), post-hoc comparisons of estimated marginal means were conducted using the emmeans package, with pairwise contrasts adjusted by Bonferroni correction [*emmeans* function from “emmeans” package (Lenth 2022)]. Among the tested variables, *nirS* (from the Introduced Pasture samples only) required additional transformation prior to the modeling. Based on the models, the p -values from the individual factors (grazing, season, and year) as well as their interactions are presented in Table S3. To evaluate the contributions of random and fixed effects to the overall variability of the tested variable, marginal R^2 , conditional R^2 , ΔR^2 (marginal—conditional), and intraclass correlation coefficient (ICC) were estimated [*r2_nakagawa* and *icc* functions from “performance” package (Lüdtke et al. 2019)] and presented in Table S4. To represent the variables according to the significance of the interactions, we applied two formats: non significant variance and three-way interactions, the variable was plotted over the sampling times, and the grazing effect was highlighted; and variables that presented significant two-way interaction (and non-significant three-way interaction) were presented in tables. Some gene abundances were presented on different y-axis scales across sampling periods due to differences in magnitude; as a result, lines are not connected between periods. The individual main effects were not discussed when three- or two-way interactions were significant. Although the Season $\times$ Year interaction was tested, those results are not presented since the primary focus is on grazing treatment effects.

Distance-based redundancy analysis (dbRDA) was performed to test the relationship between gene abundances, soil biochemical parameters, and stocking density using vegan package (Oksanen et al. 2025). Bray–Curtis dissimilarities were calculated from the gene abundance matrix; a stepwise selection was applied to select a subset of parameters. The final dbRDA model retained the subset of predictors that significantly explained variation in gene composition, while controlling temporal and soil moisture effects. Model significance was assessed using permutation ANOVA (overall and by term, number of permutations = 999), and variance inflation factors (vif.cca)

were examined to diagnose multicollinearity among retained predictors. The association between soil biochemical parameters, stocking density, and enzyme activities (C-, N-, and P-enzymes) were analyzed using Bayesian multilevel generalized linear mixed models, family Gaussian [*brm* function from “brms” package (Bürkner 2015)], to account for temporal and spatial dependencies. Each enzyme category was modeled separately as a gamma-distributed response with a log link to accommodate strictly positive and right-skewed values. Fixed effects included standardized soil predictors selected based on correlation and collinearity analyses. Random intercepts were included for sampling time and pasture to account for temporal and spatial variation. Weakly informative priors were applied, including a horseshoe prior for fixed effects to encourage sparsity, normal priors for the intercepts, and exponential priors for the standard deviation of random effects. Models were fit using four Markov chains, and convergence was assessed through the potential scale reduction factor and effective sample size. Model adequacy was further evaluated using posterior predictive checks, including comparisons of observed versus simulated distributions, posterior predictive residuals, and the coverage of credible intervals.

Results

N-cycling genes and enzyme activities in Native Rangeland

Grazing, season, year, and their interactions significantly affected most N-cycling genes, enzyme activities and enzyme stoichiometry (Table S3). The relatively small difference between conditional and marginal R^2 values suggests that most of the variance was explained by the fixed effects (Table S4). Interaction effects varied among genes; a three-way interaction (Grazing $\times$ Season $\times$ Year) was detected for the denitrification genes *napA*, *narG*, and *nirS* (Fig. 1). The abundance of the *napA* gene declined over time in both grazing treatments, with AD consistently exhibiting greater values than PR across several sampling periods (Fig. 1a). Adaptive grazing resulted in 13%, 57%, and 25% greater *napA* expression in summer 2022, fall 2022, and summer 2024, respectively, relative to prescriptive ($p < 0.05$). The abundance of

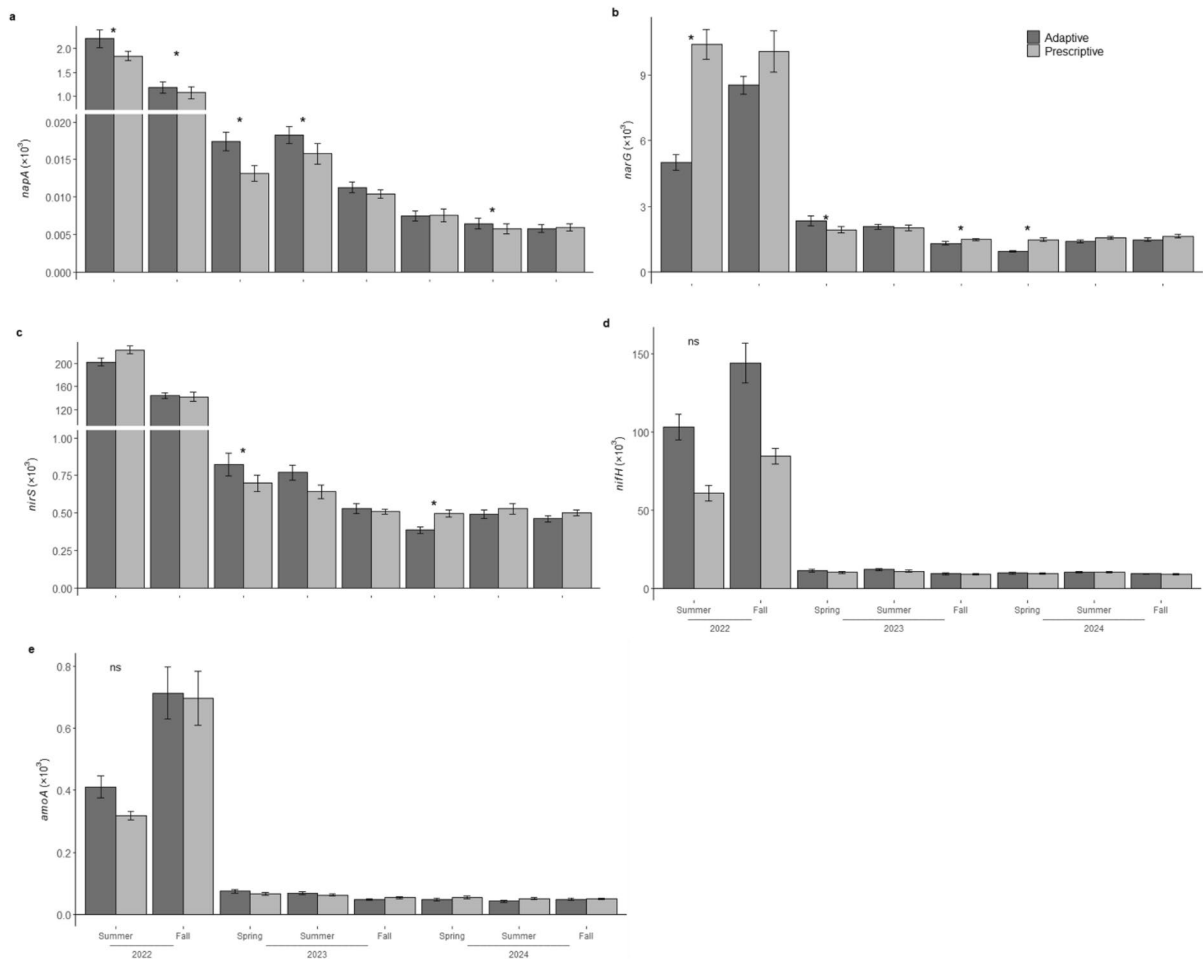


Fig. 1 Observed N-cycling genes abundance in Native Rangeland showing Grazing $\times$ Season $\times$ Year interactions; (a) *napA*, (b) *narG*, (c) *nirS*, (d) *nifH*, (e) *amoA* (normalized to 16S $\times$ 1000). Error bars indicate $\pm$ standard error. Asterisks indicate

significant differences ($p < 0.05$) between grazing treatments at a given sampling time and ns represents genes that did not present statistical significance for the three-way interaction

narG also declined over time, but in contrast to *napA*, it was generally greater under PR than AD. Significant differences in *narG* were detected in summer 2022, spring and fall of 2023, and spring 2024 (Fig. 1b, $p < 0.05$). In summer 2022, *narG* abundance of PR was twofold greater than AD, and although the difference between the treatments decreased over time, PR remained consistently greater than AD, except in spring 2023, when AD exceeded PR by 21%. The abundance of *nirS* also declined over time, with significant grazing treatment effects in spring of 2023 and spring 2024 (Fig. 1c, $p < 0.05$). In 2023, *nirS* was 18% greater in AD than PR, while by spring 2024, it was 22% lower under AD, indicating a temporal

shift in treatment response as overall abundance levels remained low. Genes associated with N fixation (*nifH*) and ammonia oxidation (*amoA*) showed no significant Grazing $\times$ Season $\times$ Year interaction nor grazing treatment effects (Fig. 1d and e, $p > 0.05$), but it's noticeable drastic decrease of measures from 2022 to the subsequent years.

Two-way interaction (Grazing $\times$ Season) was detected for the other two denitrification genes *nirK* and *nosZ* (Table 1). *NirK* abundance was lowest in spring, with no difference between summer and fall for both grazing treatments. Within-season, comparisons showed that during summer, *nirK* abundance was approximately 8% greater under AD than PR

Table 1 N-cycling gene abundance (normalized to 16S × 1000) for Grazing × Season interactions in Native Rangeland

Variable	Grazing	Season		
		Spring	Summer	Fall
<i>nirK</i>	Adaptive	495Ba	1803Aa	1860Aa
	Prescriptive	504Ba	1674Ab	1697Aa
<i>nosZ</i>	Adaptive	1.472Aa	5.039Aa	2.899Ab
	Prescriptive	1.485Aa	5.359Aa	3.907Aa

Means followed by different lowercase letters within a column indicate significant differences among grazing treatments within a given season, whereas means followed by different uppercase letters within a row indicate significant differences among seasons within a given grazing treatment ($p < 0.05$)

($p < 0.05$), while no grazing effect was detected in spring or fall. On the other hand, *nosZ* abundance was 35% greater under PR than AD only in fall. In addition, there was a significant Grazing × Year interaction for *nosZ*. Both grazing treatments exhibited the greatest abundance in 2022 (AD = 9.2; PR = 11.3), decreasing by nearly an order of magnitude by 2023 (AD = 1.4; PR = 1.5; $p < 0.05$) and remaining low through 2024 (AD 1.5; PR = 1.5; $p < 0.05$). In 2022, *nosZ* was 22% greater under PR compared to AD ($p < 0.05$), but no differences were observed in subsequent years.

Temporal patterns were evident across all enzyme groups, with significant three-way interaction among grazing, season, and year (Fig. 2). C-enzyme activity was greater under AD grazing during summer 2022 (175%), summer 2023 (35%), and spring 2024 (39%) compared to PR. In contrast, PR exhibited greater activity in fall 2022 (26%) and a pronounced peak in spring 2023 (425%) compared to AD (Fig. 2a). N-enzyme activity was 271% greater in AD than PR in summer 2022, 78% greater in summer 2023, and 55% greater in spring 2024 (Fig. 2b). P-enzyme activity was 216% greater in AD than PR in summer 2022. However, the activity declined in AD in fall 2022, becoming 30% lower than PR, and thereafter enzyme activities stabilized at low levels with no further differences detected (Fig. 2c).

The enzyme stoichiometric ratios varied over time, with alternating grazing effects across years. AD showed lower C:N ratios in fall 2022 (−10%) and summer 2023 (−14%) compared to PR, suggesting proportionally greater N-enzyme activity, whereas PR was 36% greater than AD in spring 2023 (Fig. 2d).

By fall 2023, the pattern reversed, with AD 10% greater than PR, after which ratios stabilized around 1.0–1.3 across both grazing treatments. The C:P ratio showed consistent yet shifting differences between AD and PR (Fig. 2e). In spring 2023, PR had a 43% greater ration than AD, while from late 2023 through 2024, AD surpassed PR by 8–10%. The N:P ratio was greater under AD grazing across most sampling periods (Fig. 2f). AD exceeded PR by 6–16% in summer and fall of 2022, 22% in summer 2023, and 14% in spring 2024; ratios subsequently stabilized between 0.7 and 0.9.

Distance-based redundancy analysis (dbRDA) was performed to identify linear correlations between soil biochemical parameters, stocking density, and N-cycling genes (Fig. S1a). From the tested parameters, pH, SOM, dissolved organic C (DOC), dissolved organic to inorganic N ratio (ON:IN), and stocking density ($p < 0.05$) were selected. Temporal and GWC variation explained approximately 76% of the total variance, while the predictors accounted for 1.5% of variance, and the residual explanation was about 22%. When testing predictors for enzymes activity, pH, soil respiration, SOM, C:N, IN, OP, stocking density, and GWC were selected (Fig. S1b). From these, soil respiration (0.13) presented significant estimates for C-enzymes model; and stocking density was the only predictor with significant estimates for N- and P-enzymes (0.13 and 0.18 respectively).

N-cycling genes and enzyme activities in Introduced Pasture

Grazing treatment, season, and year had significant effects on *nirS* and *nirK* gene abundance, as well as N-enzymes and stoichiometric ratios (Table S3) and explained the variance observed (Table S4). The abundance of *nirS* consistently differed between grazing treatments, remaining greater under PR across all sampling periods. Although grazing treatment differences were small in 2022, by spring 2023 AD was 38% lower than PR, and this pattern persisted through 2024, with differences ranging from 9 to 57%. There was a sharp decline in *nirS* abundance in spring 2023 in both grazing treatments, stabilizing at low levels thereafter (Fig. 3a). The abundance of *nirK* showed grazing differences limited to the early and late stages of the study (Fig. 3b). In summer 2022, AD exhibited twice the abundance of PR, but this pattern reversed

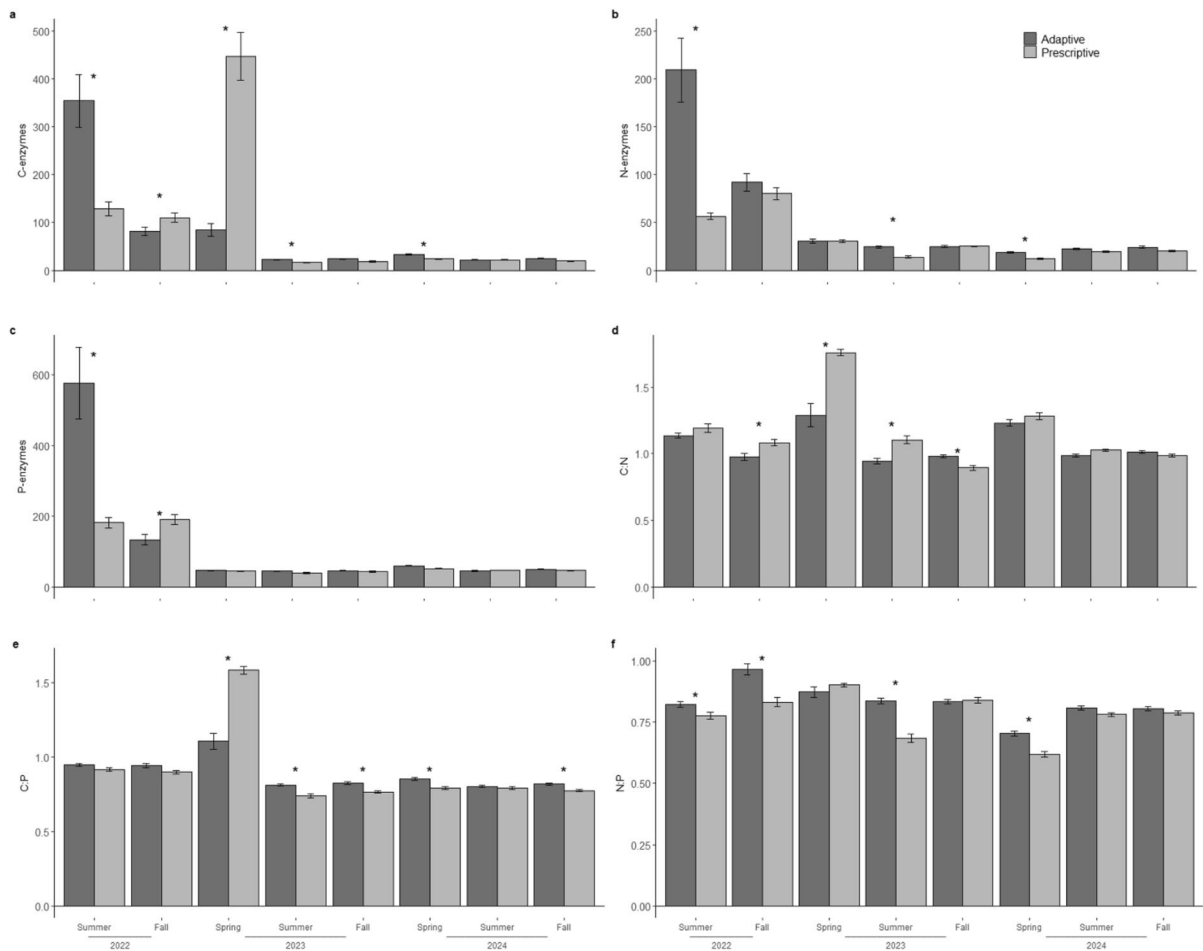


Fig. 2 Observed C- (a), N- (b), and P- (c) enzyme activity ($\text{nmol g}^{-1} \text{h}^{-1}$) and stoichiometric ratios C:N (d), C:P (e), and N:P (f) in Native Rangeland showing Grazing $\times$ Season $\times$

Year interactions. Error bars indicate $\pm$ standard error. Asterisks indicate significant differences ($p < 0.05$) between grazing treatments at a given sampling time

by spring 2024, when AD was 56% lower than PR. Despite seasonal fluctuations in between, both grazing treatments followed a similar declining trend, with *nirK* stabilizing at low levels by the end of the study. Other genes associated with N fixation (*nifH*) and denitrification (*narG* and *nosZ*) showed no significant Grazing $\times$ Season $\times$ Year interactions or the main effect of grazing treatments (Fig. 3c, d, and e, $p > 0.05$).

The abundance of *amoA* and *napA* showed a significant Grazing $\times$ Year interaction; *amoA* differed across years, with AD pastures showing 65–70% lower values than PR in 2023, whereas 2022 and 2024 showed no significant differences between grazing treatments (Table 2). In AD, *amoA* abundance

decreased fourfold from 2022 to 2023 and 2024, while in PR it decreased by 31% from 2022 to 2023, and then threefold from 2023 to 2024. Similarly, *napA* in AD declined about two orders of magnitude from 2022 to 2024; in PR the change was approximately three orders of magnitude in the same period (Table 2).

C-enzymes activity was not affected by the Grazing $\times$ Season $\times$ Year interaction (Fig. 4a), whereas N-enzymes and C, N, and P stoichiometric ratios were significantly influenced by the three-way interaction. For N-enzyme activity (Fig. 4b), PR exhibited nearly twice the activity of AD in fall 2022, remaining 21–24% greater through summer and fall 2023 and 16% greater in spring 2024. However, this

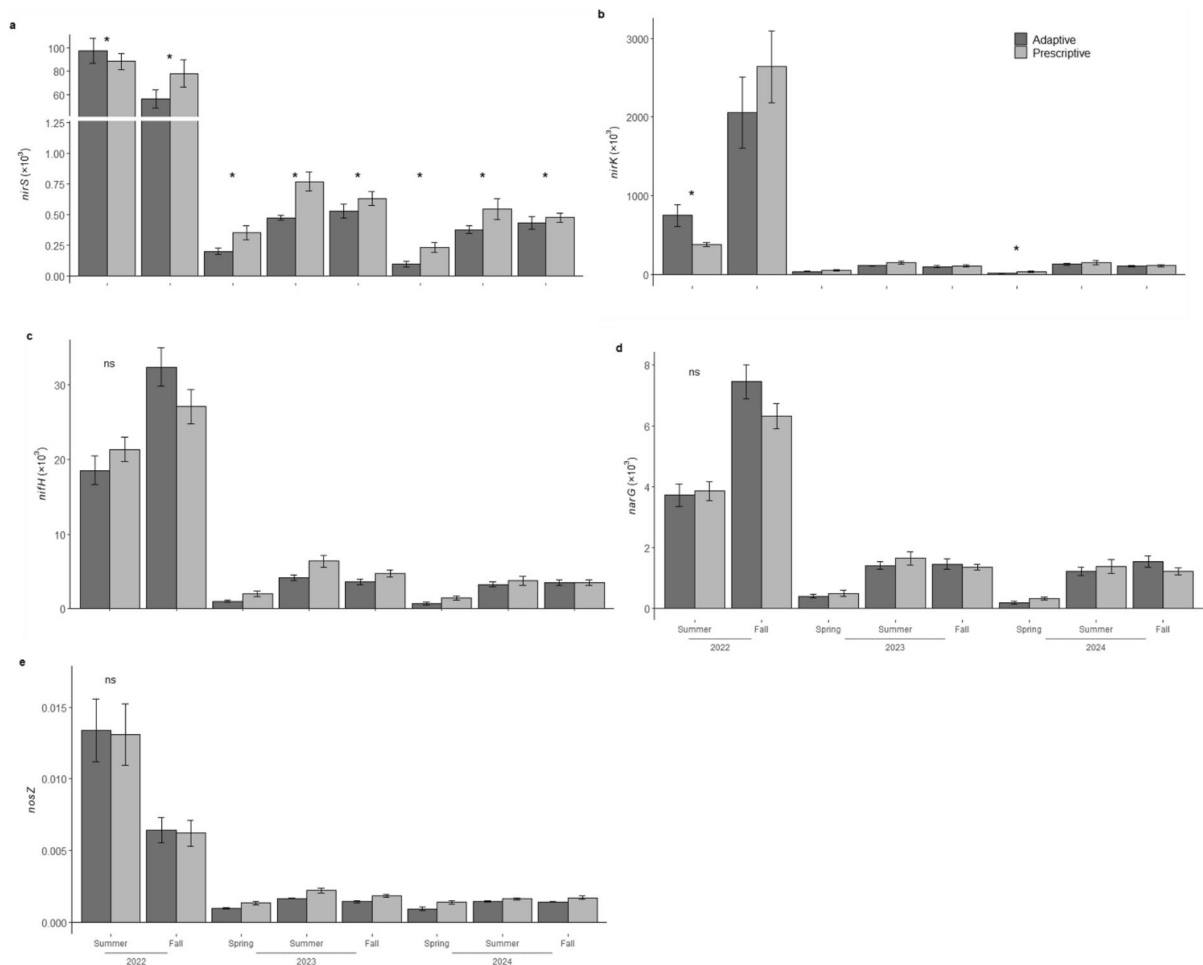


Fig. 3 Observed N-cycling gene abundance in Introduced Pasture showing Grazing × Season × Year interactions; (a) *nirS*, (b) *nirK*, (c) *nifH*, (d) *narG*, (e) *nosZ* (normalized to 16S × 1000). Error bars indicate ± standard error. Asterisks indicate

significant differences ($p < 0.05$) between grazing treatments at a given sampling time and ns represents genes that did not present statistical significance for the three-way interaction

pattern reversed by summer 2024, with AD exceeding PR by 18%. The C:N enzyme ratio remained stable over time (0.96–1.40), with a significant effect observed in fall 2023, when AD showed 9% greater ratio than PR (Fig. 4c). The C:P ratio showed grazing treatment differences in fall 2022 and summer 2024 (Fig. 4d), where AD was 15% lower than PR in fall 2022, but 17% greater by summer 2024, showing a shift toward enhanced C-acquisition in AD later in the study. Ratios remained below 1.0 overall, reflecting a sustained emphasis on P-mineralizing enzymes in both grazing treatments. Across sampling periods, N:P ratio ranged from 0.52 to 0.69 under AD and from 0.47 to 0.73 under PR, with significant grazing

treatment effects also in fall 2022 and summer 2024 (Fig. 4e). Similar to that observed for the C:P ratio, N:P ratio was 16% lower in AD in fall 2022 but 10% greater in summer 2024 compared to PR. Ratios across both grazing treatments remained below 0.8, suggesting a consistent predominance of P- over N-enzyme activity throughout the study period.

P-acquiring enzymes exhibited a significant Grazing × Year interaction, being 68% greater under PR than AD in 2022, with no differences detected in subsequent years ($p > 0.05$, Table 2). From 2022 to 2023 and 2024, P-enzymes decreased by 26% in AD and by half in PR. C-acquiring enzyme activity showed no significant interaction between grazing and time

Table 2 N-cycling gene abundance (proportion to $16S \times 1000$) and enzyme activity ($\text{nmol g}^{-1} \text{h}^{-1}$) for Grazing $\times$ Year interactions in Introduced Pasture

Variable	Grazing	Year		
		2022	2023	2024
<i>amoA</i>	Adaptive	0.733Aa	0.160Bb	0.173Ba
	Prescriptive	0.722Aa	0.549Ba	0.156Ca
<i>napA</i>	Adaptive	1.658Ab	0.008Ba	0.005Ba
	Prescriptive	2.356Aa	0.007Ba	0.007Ba
P-enzymes	Adaptive	47.96Ab	43.47Ba	32.68Ba
	Prescriptive	80.47Aa	44.63Ba	34.53Ba

Means followed by different lowercase letters within a column indicate significant differences among grazing treatments within a given year, whereas means followed by different uppercase letters within a row indicate significant differences among years within a given grazing treatment ($p < 0.05$)

[season and years (Fig. 3a, $p > 0.05$)], but it varied seasonally, with AD lower than PR in spring (AD = 17.19; PR = 26.58; $p < 0.05$) and fall (AD = 10.46; PR = 14.52; $p < 0.05$) but similar in summer (AD = 14.43; PR = 20.39; $p > 0.05$). In AD, spring values were 38% greater than summer and fall ($p < 0.05$); while in PR, C-enzyme activity decreased 30% from spring to summer ($p < 0.05$), and then 40% from summer to fall ($p < 0.05$).

The selected predictors for the dbRDA were SOM, IN, IP, and OP ($p < 0.05$, Fig. S2a). Conditioned parameters (temporal and GWC variation) explained 50% of the overall variance, predictors accounted for 2%, leaving 48% unconstrained variance. The individual contribution of soil biochemical parameters and stocking density on enzyme variation over time was tested through regression analysis (Fig. S2b). From the selected parameters (pH, soil respiration, SOM, C:N, IN, OP, stocking density, and GWC), SOM (0.21) and C:N (0.12) presented significant estimates for C-enzymes; soil respiration (0.09) for N-enzymes; and SOM (0.13) and C:N (0.1) for P-enzymes.

Discussion

This study evaluated the impact of grazing management on N-cycling genes and C-, N-, P- acquiring enzymes over three years in two contrasting pasture systems: Native Rangeland and Introduced Pasture. We hypothesized that AD grazing, characterized by

greater stocking densities and longer rest periods, would enhance microbial functioning compared to PR (lower stocking densities with fixed rest periods during the growing season). Our results revealed complex and system-specific responses. In Native Rangeland, AD generally promoted greater microbial gene abundance and enzyme activity, particularly during spring and summer seasons, with notable increases in *napA* and N-enzymes, compared to PR. However, some genes (e.g., *nirS*) showed shifts over time, with greater abundance in AD than PR in early stages of the study, but the opposite in subsequent years. In Introduced Pasture, PR supported greater *nirS* abundance and N-enzyme activity. Enzyme stoichiometric ratios further highlighted shifts in microbial nutrient acquisition strategies, with AD favoring N and P acquisition in Native Rangeland and showing late-stage improvements in C and P acquisition in Introduced Pasture. These findings underscore the importance of land-use, seasonal dynamics, and long-term monitoring in evaluating grazing impacts on soil microbial processes.

From summer 2022 to fall 2024, we observed an overall decline in the abundance of N-cycling genes and enzyme activity. During this period, the studied area was under severe drought for several consecutive months, mostly during 2022 and 2023 (Sparks et al. 2026). For example, from summer 2022 through fall 2024, summers presented rainfall averages 123% lower than historical average, while spring 2023 was 15% lower, and falls were close to historical averages. While microbial functionality was high in 2022, it dropped significantly in 2023 and remained low through 2024. This pattern aligns with recent findings suggesting that drought-induced N_2O emissions follow an inverted U-shaped response curve post rewetting (Barrat et al. 2022), where moderate droughts can stimulate microbial activity and N_2O emissions, but prolonged droughts suppress microbial function due to stress-induced cell death and lysis (Clark and Hirsch 2008; Meisner et al. 2015; Schimel 2018). Microbial community may utilize a two-stage dormancy strategy, storing osmolytes during dry periods and catabolizing them upon rewetting. However, at prolonged negative water potentials this strategy is no longer effective, and microbes enter a deeper state of dormancy, resulting in a community that can no longer respond rapidly to changes in water availability (Barrat et al. 2022). The lack of recovery in gene

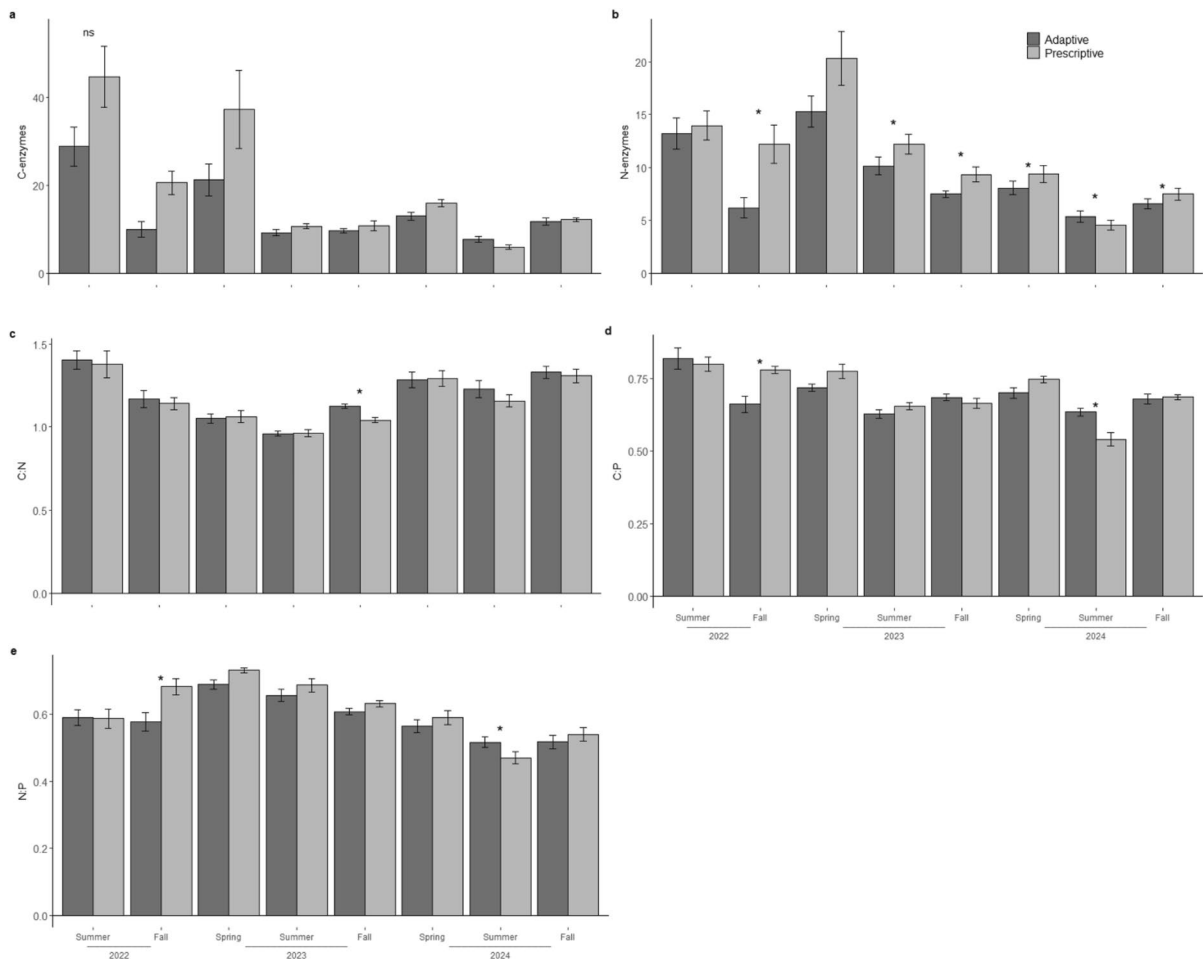


Fig. 4 Observed C- (a) and N- (b) enzyme activity in $\text{nmol g}^{-1} \text{h}^{-1}$, and stoichiometric ratios C:N (c), C:P (d), and N:P (e) in Introduced Pasture showing Grazing $\times$ Season $\times$ Year interactions. Error bars indicate $\pm$ standard error. Asterisks indicate

significant differences ($p < 0.05$) between grazing treatments at a given sampling time and ns represents enzymes that did not present statistical significance for the three-way interaction

abundance in 2024 may reflect cumulative microbial stress and reduced community resilience, showing diminished microbial functionality post-drought.

Changes in N-genes abundance under grazing

Significant differences between AD and PR were more commonly observed in the dissimilatory nitrate reduction and denitrification pathway (*napA*, *narG*, *nirK*, and *nirS*). *NapA* encodes the periplasmic nitrate reductase, an enzyme involved in the dissimilatory nitrate reduction pathway that ultimately generates NH_4^+ (Asamoto et al. 2021; He et al. 2025). In this study, *napA* abundance was consistently greater under

AD than PR in Native Rangeland. This pattern may reflect the more dynamic redox environment under AD grazing, where greater root and litter inputs stimulate microbial respiration and transient oxygen depletion (Chen et al. 2016; Zhou et al. 2017). Under such conditions, *napA*-type nitrate reduction can operate as nitrate scavenger in a redox-balancing mechanism in both aerobic and micro-anaerobic environments (Asamoto et al. 2021). Ellington et al. (2002) demonstrated that *napA* abundance increases when reduced C sourced creates a redox imbalance, allowing cells to use *nap* pathway as an alternative electron sink under limited oxygen conditions. Wu et al. (2021) reported that organic amendments such

as manure and peat reduced *napA* abundance, further suggesting that moderate C availability and fluctuating oxygen conditions may favor *napA*-mediated nitrate reduction.

NarG plays a role in the initial denitrification step by converting nitrate (NO_3^-) into nitrite (NO_2^-) (Li et al. 2025) and its abundance has been correlated with NO_3^- content (Jia et al. 2025). The greater *narG* abundance in PR, particularly during fall and spring sampling periods, may indicate greater NO_3^- availability in these soils. Reinforcing this assumption, N fractions were predictors of gene abundance in the dbRDA analysis (Figs. S1 and S2). If considering that animals can represent an input of up to 1000 kg ha^{-1} of N (Selbie et al. 2015), timing and frequency of grazing events influence N accumulation in the soil. According to Gourley et al. (2025), when urine is deposited in late spring, plants may not fully utilize nitrogen before soil moisture becomes limiting, reducing N uptake, and increasing nitrate accumulation in both surface and subsurface soils. During the study, PR pastures in Native Rangeland were grazed eight times, against five grazing events in AD. PR grazing occurred consecutively once per season during spring, summer, and fall, with occasional grazing in winter. Under AD, grazing events occurred at longer intervals, with variation in the timing of the grazing events; primarily occurring in spring and summer, with approximately half as many events in fall compared to PR, while winter events occurred at a similar frequency between the two grazing treatments (Sparks et al. 2026). Greater and consistent grazing frequency in PR may have led to more frequent nutrient inputs, particularly during limited plant uptake periods, potentially contributing to elevated nitrate levels and stimulating *narG* abundance. Corroboratively, stocking density was a significant predictor of variance in gene abundance in Native Rangeland.

The *nirK* and *nirS* genes are commonly used as markers to quantify denitrifying microorganisms (Ouyang et al. 2018; Song et al. 2020) and are sensitive to agricultural management. Although both encode functionally similar nitrate reductases, differences in their abundance reflect distinct habitat preferences and functional traits, suggesting that *nirK*- and *nirS*-type denitrifiers occupy different ecological niches (Enwall et al. 2010). In this study, *nirS* showed significant grazing treatment effect in both ranches, whereas *nirK* only showed significance in Introduced

Pasture, indicating that the response of these genes to grazing was site-specific. Within Introduced Pasture, *nirS* was consistently greater under PR than AD, while responses to grazing in the Native Rangeland were inconsistent. In contrast, *nirK* showed sparse and inconsistent grazing treatment effects. These results align with previous findings showing that edaphic factors can shape the distribution and dynamics of denitrifying communities (Enwall et al. 2010; Yang et al. 2025).

White et al. (2023) reported that *nirK* was 113% more abundant under high stocking density grazing ($> 60 \text{ AU ha}^{-1}$; rest/grazed ratio > 40 days) compared to conventional grazing, characterized by low to moderate stocking densities with infrequent movement of cattle among paddocks. In contrast, *nirK* responses in our study were less consistent. In Native Rangeland, *nirK* abundance in AD was 8% greater than PR only in summer. In Introduced Pasture, *nirK* was greater in AD than PR only in summer 2022, with a reversed effect in spring 2024. For *nirS*, grazing effects were minimal in Native Rangeland, whereas in Introduced Pasture, *nirS* was consistently greater under PR than AD, particularly in 2023 and 2024. This pattern may be linked to greater grazing events frequency associated with lower stocking density [5 AU ha^{-1} vs 35 AU ha^{-1} (Sparks et al. 2026)] observed in PR in Introduced Pasture, potentially allowing for greater plant biomass accumulation (Kotzé et al. 2017) and organic matter inputs. Such conditions can enhance soil C and N availability, thereby stimulating denitrification activity. González-Coria et al. (2025) similarly observed that denitrifying microbes rely on C and N substrates, and that microbial respiration stimulated by phenolic compounds can release recalcitrant C, which further supports denitrification processes. Additionally, the observed contrasting responses between *nirK* and *nirS* suggest that the two denitrifier groups may occupy distinct ecological niches or differ in their sensitivity to substrate availability, as previously proposed by Braker et al. (1998), Enwall et al. (2010), and Yang et al. (2021).

Enwall et al. (2010) found that soil nitrate, DOC, and clay content significantly influenced the spatial distribution of *nirS* communities. Similarly, Yang et al. (2025) reported that the soil texture affected the temporal response of both *nirK* and *nirS* genes to management practices. In our study, differences in soil texture [finer in Native Rangeland and coarser in

Introduced Pasture (Sparks et al. 2026)] may partly explain the site-specific responses observed for *nirK* and *nirS*. Additionally, SOM and N fractions were significant predictors for the variance in N-cycling gene abundance in both Native Rangeland and Introduced Pasture.

Among the seven tested genes, *nifH* was the only one that showed no grazing treatment effect, either as two- or three-way interactions, in either location. This lack of response suggests that, under the conditions of this study, N-fixation potential was less responsive to grazing-induced changes than nitrate reduction and denitrification pathways. Previous work indicates that *nifH* responses to herbivory can depend strongly on the biological compartment evaluated. In a Serengeti herbivore exclusion experiment, exclusion increased *nifH* copy number in roots of the legume *Indigofera volkensii*, but did not affect *nifH* copy number in roots of the grass *Themeda triandra* or in soil (Ritchie and Raina 2016). This is consistent with our results, because our measurements were based on bulk soil rather than host-specific root associations, where grazing effects on diazotrophs may be more detectable. In addition, Mitsuta et al. (2026) showed that *nifH* abundance in an acidic grassland meadow was shaped primarily by soil pH changes induced by long-term fertilizer application, rather than by short-term nutrient inputs. In the present study, soils were close to neutral pH, which may have reduced pH-related constraints on diazotroph abundance and contributed to the absence of a grazing response. Together, these findings suggest that *nifH* may be more strongly regulated by host identity, soil pH, and broader edaphic conditions than by the grazing treatments evaluated in this study.

Changes in enzyme activity and stoichiometry under grazing

Herbivore grazing alters plant tissue stoichiometry, with subsequent effects on soil nutrient inputs and microbial processes. Previous studies have shown that grazing may enhance N concentrations in plant tissues, although responses are often species-specific (Bai et al. 2012; Yang et al. 2017; Ma et al. 2019). Since leaf, stem, and root stoichiometry are closely linked to soil C, N, and P pools (Ordoñez et al. 2009; Zheng et al. 2012), shifts in plant allocation can cascade to the soil microbial community.

Zhang et al. (2025) showed that the dung deposition accelerates soil nutrient cycling, mainly by shifting soil pH and moisture, altering the composition of bacterial communities and ultimately enhancing the enzymatic activities. In our study, enzyme activity and stoichiometric ratios exhibited significant interactions with grazing, season and year, but the direction of these responses varied among systems.

Both LAP and NAG are crucial enzymes responsible for N acquisition through the degradation of proteins and chitin (Daunoras et al. 2024), and their activities were combined to represent the N-enzymes in this study. The activity of N-enzymes is typically enhanced under N-limited conditions, as microbes increase enzyme production to access N sources (Uwituze et al. 2022). In Introduced Pasture, PR presented greater N-enzyme activity than AD, suggesting lower N availability in PR soils. Greater *nirS* abundance under PR may reflect enhanced denitrification potential driven by higher soil C availability, which can further deplete inorganic N pools and reinforce microbial N limitation. Also, soil respiration had a positive relationship with the changes in N-enzymes over time. These results suggest that PR grazing fosters conditions of elevated C availability but lower N accessibility, leading to increased microbial demand for N-acquiring enzymes (Wang et al. 2020).

The use of enzyme stoichiometry has been proposed as an effective approach to infer microbial nutrient limitations and resource allocation strategies (Liu et al. 2020). Based on a global-scale meta-analysis of the seven-most widely measured soil enzyme activities across 40 ecosystems, Sinsabaugh et al. (2008) concluded that microbial C:N:P acquisition ratios converged towards a 1:1:1 balance, reflecting coordinated nutrient cycling. In Native Rangeland, where stoichiometric ratios were close to 1.0, AD consistently exhibited greater C:P and N:P ratios compared to PR. This pattern suggests that the greater stocking density (76 AU ha⁻¹) and longer rest period (121 days) under AD enhanced microbial energy allocation toward C and N acquisition, likely due to increased substrate inputs and turnover. Temporal trends further indicate a potential cumulative grazing effect—for C:N ratio, PR initially exceeded AD, with no differences at the end of the study; for C:P ratio, PR peaked in spring 2023, after which AD consistently exceeded PR from summer 2023 onward; and

for N:P ratio, PR showed greater seasonal variability, whereas AD remain consistently greater than PR.

In Introduced Pasture, enzyme stoichiometric ratios revealed no consistent differences between grazing treatments but indicated persistent nutrient limitation across the system. It is important to note that this system had not received fertilizer inputs for the previous four years. C:N ratio remained near or above 1.0 under both grazing treatments, suggesting balanced microbial investment in C and N acquisition, with a slight preference toward C. In contrast, C:P and N:P ratios consistently remained below 0.8, reflecting a sustained predominance of P-mineralizing enzymes and a strong microbial demand for P. This pattern aligns with our dbRDA results, that revealed a significant association between soil environmental characteristics and gene community composition after accounting for sampling time and soil water content (totaling 87% of gene variance explained), in which P fractions were consistently selected as predictors of N-cycling gene abundance, indicating a relationship between P availability and microbial functionality.

Nitrogen availability can stimulate microbial function and increase demand for C and P, prompting microbes to produce additional enzymes to hydrolyze labile SOM (Yang et al. 2022), and in our study temporal shifts in P-enzymes were associated with SOM and soil C:N dynamics. Zhao et al. (2025) reported that grazing can shift microbial metabolic limitations depending on grazing intensity and ecosystem type, with P limitation often prevailing under low grazing pressure and transitions toward N or C limitation emerging as grazing intensity increases. The dominance of P-acquisition in our Introduced Pasture suggests that, despite grazing-mediated nutrient returns, microbial processes remain constrained by P availability. Overall, these findings reinforce that grazing effects on microbial metabolism are context-dependent and shaped by site-specific properties and the balance among nutrient inputs, removals, and recycling.

The minimal differences in enzymatic stoichiometry observed between grazing treatments in Introduced Pasture suggest that grazing effects were weak at the time of sampling. This limited response is likely linked to the temporal sensitivity of enzyme activities, as samples were collected 15 days after grazing. Because enzyme dynamics can fluctuate over short time scales, our measurements may not have captured transient shifts immediately following livestock

movement. Prior work has shown strong temporal fluxes in enzyme activity within days to weeks after grazing, regardless of grazing intensity or duration (Van Syoc et al. 2022), and enzymatic processes are known to exhibit rapid variability in response to environmental and abiotic conditions (Baldrian 2014; Boeddinghaus et al. 2015).

The absence of significant estimates for enzyme activity and stocking density can also be explained by minimal contrast between grazing strategies, potentially reflecting the moderate difference in grazing intensity (35 AU ha⁻¹ in AD versus 5 AU ha⁻¹ in PR). An additional explanation is that microbial activity was driven primarily by seasonal and interannual variability, with grazing effects secondary to broader environmental drivers. A review of nearly 300 studies by Daunoras et al. (2024) reported that soil temperature, moisture, substrate availability, climatic factors such as average annual temperature and precipitation strongly influence microbial processes. Likewise, a meta-analysis by Brzostek et al. (2012) showed that potential proteolytic activity (an important step in N cycling) declines under soil moisture deficits (the difference between precipitation and evapotranspiration) in dry grasslands. Thus, the severe drought conditions observed during our study likely constrained enzyme activity, functioning as a limiting factor. Collectively, these findings emphasize the need for finer temporal resolution in longitudinal grazing research. Future sampling strategies should incorporate shorter and repeated intervals immediately following grazing events to resolve short-term microbial responses and better distinguish management effects from overarching environmental variability.

Our results highlight how the interaction of stocking density and recovery time mediates microbial nutrient acquisition strategies in a context-dependent manner, reinforcing the need to account for site-specific conditions when applying sustainable grazing strategies (Liu et al. 2024). Overall, our findings demonstrate that the effects of AD and PR on soil microbial functioning diverge between Native and Introduced systems, reflecting differences in plant communities, litter dynamics, and nutrient cycling pathways. However, seasonal and interannual variability strongly modulated microbial functioning in both systems, with drought years (2022–2023) coinciding with subsequent reduced gene abundance and enzyme activity (2023–2024). This pattern is

consistent with our recent work, where treatment-associated shifts in dominant bacterial phyla could not be attributed to grazing management alone but instead appeared to reflect the combined influence of recurring drought–rewetting cycles and grazing-associated environmental filtering (Chi et al. 2026). Together, these findings reinforce that microbial functioning is jointly controlled by grazing management and climatic conditions. Similarly, Jia et al. (2025) reported that microbial functional genes were highly sensitive to fertilization in a long-term wheat–maize system, proposing their use as indicators of soil health. While our results support the potential use of N-cycling genes and nutrient-acquiring enzymes as indicators of management impact on soil health, the high sensitivity of these parameters to environmental variability suggests that replication and climatic context are critical for robust interpretation.

Conclusions

We monitored N-cycling genes and C-, N-, and P-acquiring enzymes under two grazing strategies: AD (higher stocking density with longer rest periods) and PR (lower stocking density with fixed rest periods). By examining contrasting systems (Native Rangeland and Introduced Pasture), microbial responses to grazing and the influence of site-specific conditions were evaluated. Our hypothesis that grazing management would alter N-cycling gene abundance was partially supported. Effects were most evident in genes involved in dissimilatory nitrate reduction and denitrification (*napA*, *narG*, *nirK*, *nirS*), indicating shifts in N availability, but responses were both gene- and site-dependent. These contrasting responses among genes suggest that the denitrifier groups occupy distinct ecological niches or differ in sensitivity to substrate availability. Across models, SOM, N and P fractions were significant predictors of gene abundance, linking substrate availability to microbial functional responses. Enzyme activity patterns reinforced these findings. N-enzymes showed the most consistent grazing response across both systems, with greater activity under AD in Native Rangeland and under PR in Introduced Pasture. Stoichiometric ratios close to 1.0 in Native Rangeland indicated balanced microbial investment in nutrient acquisition. In contrast, C:P and N:P ratios remained

below 0.8 in Introduced Pasture, reflecting predominant P-mineralizing activity and sustained microbial demand for phosphorus. Overall, our results show that N-cycling genes and nutrient-acquiring enzymes have strong potential as indicators of management effects on soil functioning. However, their high sensitivity to environmental variability presents both a challenge and an opportunity: while appropriate replication and climatic context are critical for interpretation, these microbial parameters provide valuable resolution for tracking functional responses to management.

Acknowledgements This project was funded by the Oklahoma National Science Foundation Established Program to Stimulate Competitive Research (EPSCoR; grant no. OIA-1946093). Additional support was provided by the Noble Research Institute LLC. The content of this publication is solely the responsibility of the authors and does not necessarily represent the official views of the funders.

Author contributions E.M.A.: Investigation, data curation, formal analysis, visualization, writing—original draft. I.C.F.M.: Supervision, investigation, writing—original draft, writing—review & editing. M.L.S.: Investigation, writing—review & editing. G.F.S.C.: Investigation, writing—review & editing. E.E.O.: Methodology, investigation, writing—review & editing. M.C.: Project administration, investigation, writing—review & editing.

Funding Established Program to Stimulate Competitive Research, OIA-1946093.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interests.

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